Josh Calcino:

Curriculum Vitae

Personal Information

EMAIL: jcalcino@tsinghua.edu.cn

Employment

01/01/2024 –31/01/2027 **Shuimu Scholar**
Department of Astronomy, Tsinghua University
Independent postdoctoral research on late-stage infall and substructures in planet-forming discs, combining hydrodynamic simulations with synthetic observations.

29/03/2021 –24/11/2023 **Director's Postdoctoral Fellow**
Theoretical Division, Los Alamos National Laboratory
Computational research on planet-forming discs and the dynamics of binary black holes embedded in AGN discs.

30/11/2020 –26/02/2021 **Postdoctoral Fellow**
School of Mathematics and Physics, The University of Queensland
Calibrated spectroscopic data for active galactic nuclei in a short-term postdoctoral research project.

Education

26/07/2016 –17/02/2021 **PhD in Astrophysics**
University of Queensland, Australia
Thesis: Signatures of Companions in Transition Discs: Spiral arms, cavities, and dust asymmetries. Principal supervisor: Prof. Tamara M. Davis. PhD conferred 17/02/2021, and the topic was initially in the field of cosmology.

02/03/2015 –19/07/2016 **1st Class Hons. in Astrophysics**
University of Queensland, Australia
Thesis: The effects of Redshift Bias on Cosmology. Supervisor: Prof. Tamara M. Davis.

27/02/2012 –22/11/2014 **Bachelor of Science**
Griffith University, Australia
Majors: Physics, Applied Mathematics

Selected Talks, Seminars, Posters

Overall record: 1 invited talk, 1 plenary talk, 9 contributed talks, 10 seminars, and 4 conference posters.

July 2026 **Poster**, *Companion+Infall Driven Substructures in AB Aur*, Discs on the Exe, University of Exeter, UK.

June 2026 **Invited talk**, *The Messy Leftovers of Star Formation: Morphological, Kinematic, and Chemical Signatures of Infall*, 8th Phantom Users Workshop, Macquarie University, Sydney, Australia.

May 2026 **Contributed talk**, *Click to Insert Infall: Spirals, Kinematics, and (Some) Chemistry*, Planet Formation Across Various Environments and Epochs workshop, Garching, Germany.

February 2026 **Seminar**, *Substructures in Planet-forming Discs: The Interplay of Companions, Instabilities, and Environment*, Niels Bohr Institute, Copenhagen, Denmark.

February 2026 **Seminar**, *Substructures in Planet-forming Discs: The Interplay of Companions, Instabilities, and Environment*, PFE Coffee, Max Planck Institute for Astronomy, Heidelberg, Germany.

December 2025	**Poster**, *Infall-Induced Spiral Arms and Kinematic Perturbations in Protoplanetary Disks*, Exoplanets & Planet Formation Workshop, Shanghai, China.
June 2025	**Contributed talk**, *Infall as a Source of Substructures in Protoplanetary Discs*, 7th Phantom Users Workshop, IPAG, Grenoble, France.
November 2024	**Seminar**, *Substructures in Planet-forming Discs: Companions, GI, or Something Else?*, Peking University, Beijing.
June 2024	**Seminar**, *Kinematic Signatures of Circumbinary Discs: Spiral Arms, Fast Flows, and Doppler Flips*, University of Liege, Belgium.
December 2023	**Contributed talk**, *Kinematic Signatures of Circumbinary Discs: Spiral Arms, Fast Flows, and Doppler Flips*, Exoplanets & Planet Formation Workshop 2023, Yanqing, China.
March 2023	**Contributed talk**, *Eccentric BBHs in AGN Discs: Long- and Short-Term Evolution*, AGN Santa Fe: Where are the Objects in AGN Disks?, Santa Fe, USA.
May 2022	**Plenary talk**, *The First Detection of Planetary Wakes from an Embedded Planet*, Exoplanets IV, Las Vegas, USA.
May 2019	**Poster**, *Signatures of an Eccentric Disk: Dust and Gas in IRS48*, New Horizons in Planetary Systems, Victoria, Canada. **Awarded travel support**

Public Outreach

Ongoing	**Astrophotography** *Description: I take photos of the beautiful night sky during my travels. One of my photographs has been used on the cover of Air Guitar Anthems, released by Brian May in November 2016 (I received a signed copy!).*
August 2017	**Flying Scientist** Wonder of Science, University of Queensland *Description: I flew to rural Australia accompanying the Flying Scientist for National Science Week. During my visit I gave two public talks and discussed astronomy with school students.*
March 2021	**Panellist, *Pioneers in Science*** World Science Festival Brisbane, Queensland Museum *Description: Invited panellist for a schools event at the World Science Festival Brisbane (25 March 2021), alongside Prof. Tamara Davis and Nobel Laureate Prof. Brian Schmidt, moderated by ABC science broadcaster Tegan Taylor. The event traced the paths of three researchers at different career stages, from school student to Nobel Laureate, for an audience of high-school students.*
June 2017 –July 2017	**Astronomer in Residence** ARC Centre of Excellence for All-Sky Astrophysics (CAASTRO) *Description: I was the CAASTRO Astronomer in Residence for three weeks in Yulara, a small town situated near Uluru. For three weeks I conducted public outreach, astronomy tours, and gave weekly lectures on astronomy.*
March 2017 –June 2017	**Young Science Ambassador** Wonder of Science, University of Queensland *Description: I visited schools in rural Australia for the Wonder of Science program. I ran sessions for multiple year levels, in which students addressed certain scientific challenges in astronomy and other areas of science.*

I have also filmed a segment on how black holes are detected for the national children's science programme *Scope* (Network Ten, season 4, episode 59); given six one-hour public lectures, including Luminary Lectures at Helensvale and Broadbeach Libraries for National Science Week and two at the Charleville Cosmos Centre; participated in Pint of Science Brisbane (sold out crowd); and visited roughly a dozen schools in metropolitan and rural Australia to encourage students into STEM fields.

Selected audience feedback from these public lectures: *"Impeccable presentation. Blown away. Thank you."* · *"So Amazing! Really appreciated the simple to understand language, well structured, so informative, great energy and enthusiasm."* · *"Very informative, great presenter, made subject interesting and understandable."*

Funding and Awards

- **Principal Investigator**, National Natural Science Foundation of China Research Fund for International Young Scientists, grant W2533008, **€50,000**.
- Shuimu Postdoctoral Fellow
- LANL Director's Postdoctoral Fellow
- Awarded **€3,700** in travel support during my honours and PhD.
- 2× Finalist in the University of Queensland, Faculty of Science Future Superstar Award
- NASA Astronomy Picture of the Day for September 1st, 2016 (see here)

Committees and Memberships

- Student representative on a selection committee for two astrophysics appointments, one three-year postdoctoral position and one permanent position (Jul 2018 - Nov 2018).
- Member of the ARC Centre of Excellence for All-sky Astrophysics (CAASTRO) (Feb 2015 - Mar 2018).
- Member of the CAASTRO student committee (Oct 2015 - Mar 2018).
- LOC chair and SOC co-chair for the Harley Wood School of Astronomy in 2019 (∼80 participants)
- LOC member for Australian Astronomical Society's 2019 annual meeting in Brisbane (∼250 participants)
- LOC member for the Dark Energy Survey 2016 Collaboration Meeting in Brisbane (∼150 participants)

Teaching and Supervision

- Supervised three undergraduate students on astrophysics research projects at the University of Queensland and Los Alamos National Laboratory.
- Guest lecturer for Space Science & Stellar Astrophysics, a second-year astrophysics course at the University of Queensland.
- Tutor for two consecutive years for Space Science & Stellar Astrophysics
- Laboratory demonstrator for first year physics at the University of Queensland

Publications

Total: 43 publications (42 refereed), 2396 citations, h-index=26, i10-index=39, m-index=2.6.
First author: 10 publications, 334 citations, h-index=9. Metrics from ADS, August 2026, where the full publication list is also available.

First-author publications

1. **Calcino, J.**, and 2 co-authors 2026. *Infall Explains the Disk Kinematics of AB Aur without Gravitational Instability*. Astrophysical Journal Letters, 1006, L22 (citations=5).
2. **Calcino, J.**, and 5 co-authors 2025. *Anatomy of a fall: stationary and super-Keplerian spiral arms generated by accretion streamers in protostellar discs*. Monthly Notices of the Royal Astronomical Society, 537, 2695-2707 (citations=20).
3. **Calcino, J.**, and 9 co-authors 2024. *Observational signatures of circumbinary discs - II. Kinematic signatures in velocity residuals*. Monthly Notices of the Royal Astronomical Society, 534, 2904-2917 (citations=12).
4. **Calcino, J.**, and 3 co-authors 2024. *Runaway Eccentricity Growth: A Pathway for Binary Black Hole Mergers in AGN Disks*. The Astrophysical Journal, 970, 107 (citations=30).
5. **Calcino, J.**, and 7 co-authors 2023. *Observational signatures of circumbinary discs - I. Kinematics*. Monthly Notices of the Royal Astronomical Society, 523, 5763-5788 (citations=12).

6. **Calcino, J.**, and 6 co-authors 2022. *Mapping the Planetary Wake in HD 163296 with Kinematics*. Astrophysical Journal Letters, 929, L25 (citations=34).
7. **Calcino, J.**, and 6 co-authors 2020. *Are the spiral arms in the MWC 758 protoplanetary disc driven by a companion inside the cavity?*. Monthly Notices of the Royal Astronomical Society, 498, 639-650 (citations=45).
8. **Calcino, J.**, and 7 co-authors 2019. *Signatures of an eccentric disc cavity: Dust and gas in IRS 48*. Monthly Notices of the Royal Astronomical Society, 490, 2579-2587 (citations=53).
9. **Calcino, J.**, and 2 co-authors 2018. *Updating the MACHO fraction of the Milky Way dark halo with improved mass models*. Monthly Notices of the Royal Astronomical Society, 479, 2889-2905 (citations=83).
10. **Calcino, J.**, and 1 co-author 2017. *The need for accurate redshifts in supernova cosmology*. Journal of Cosmology and Astroparticle Physics, 2017, 038 (citations=40).

Selected co-authored publications

1. Poblete, P.P., (+6 co-authors including **Calcino, J.**) 2026. *Environmental interactions in Class II systems and their impact on the disk-planet architecture*. Astronomy & Astrophysics, 712, A140 (citations=0).
2. Speedie, J., (+12 co-authors including **Calcino, J.**) 2025. *Mapping the Merging Zone of Late Infall in the AB Aur Planet-forming System*. Astrophysical Journal Letters, 981, L30 (citations=32).
3. Pinte, C., (+11 co-authors including **Calcino, J.**) 2023. *Kinematic and thermal signatures of the directly imaged protoplanet candidate around Elias 2-24*. Monthly Notices of the Royal Astronomical Society, 526, L41-L46 (citations=18).
4. Norfolk, B.J., (+11 co-authors including **Calcino, J.**) 2022. *The Origin of the Doppler Flip in HD 100546: A Large-scale Spiral Arm Generated by an Inner Binary Companion*. Astrophysical Journal Letters, 936, L4 (citations=23).
5. Verrios, H.J., Price, D.J., Pinte, C., Hilder, T., **Calcino, J.** 2022. *Kinematic Evidence for an Embedded Planet in the IM Lupi Disk*. Astrophysical Journal Letters, 934, L11 (citations=18).
6. Garg, H., (+8 co-authors including **Calcino, J.**) 2022. *A kinematic excess in the annular gap and gas-depleted cavity in the disc around HD 169142*. Monthly Notices of the Royal Astronomical Society, 517, 5942-5958 (citations=17).
7. Hammond, I., (+11 co-authors including **Calcino, J.**) 2022. *External or internal companion exciting the spiral arms in CQ Tau?*. Monthly Notices of the Royal Astronomical Society, 515, 6109-6121 (citations=14).
8. Poblete, P.P., (+14 co-authors including **Calcino, J.**) 2022. *The protoplanetary disc around HD 169142: circumstellar or circumbinary?*. Monthly Notices of the Royal Astronomical Society, 510, 205-215 (citations=11).
9. Ragusa, E., Alexander, R., **Calcino, J.**, Hirsh, K., Price, D.J. 2020. *The evolution of large cavities and disc eccentricity in circumbinary discs*. Monthly Notices of the Royal Astronomical Society, 499, 3362-3380 (citations=86).
10. Poblete, P.P., **Calcino, J.**, (+7 co-authors) 2020. *Binary-induced spiral arms inside the disc cavity of AB Aurigae*. Monthly Notices of the Royal Astronomical Society, 496, 2362-2371 (citations=33).
11. Davis, T.M., Hinton, S.R., Howlett, C., **Calcino, J.** 2019. *Can redshift errors bias measurements of the Hubble Constant?*. Monthly Notices of the Royal Astronomical Society, 490, 2948-2957 (citations=64).